

ECON601 TA Notes: Consumption

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1 Consumption Dynamics under Certainty

1.1 Standard Consumption-saving Problem

Income, Asset, Budget Constraint The consumer is endowed by a sequence of income $\{y_t\}_{t=0}^{\infty}$. In each period, he could save his income by an asset a_t at interest rate r : for each dollar saved in a_t , the consumer gets $(1+r)$ dollars in the next period. Hence, in each period $t = 0, 1, \dots, \infty$, the consumer's budget constraint is given by

$$c_t + a_{t+1} = (1+r)a_t + y_t \tag{1}$$

Note:

- The initial asset position a_0 is given.
- For now we don't put any restrictions on a_t : it can take any real numbers. $a_t > 0$ means the consumer is a saver and $a_t < 0$ means the consumer is a borrower.
- You should keep in mind that there are infinitely many such constraints (since $t = 0, 1, \dots, \infty$).

Preference The consumer's objective is his lifetime utility

$$U(c_0, c_1, \dots, c_\infty) = \sum_{t=0}^{\infty} \beta^t u(c_t) \quad (2)$$

where $\beta \in (0, 1)$ is the discount factor and $u(\cdot)$ is the period utility function. Assume that $u(\cdot)$ is increasing (the more the better) and concave (consumption smoothing motive).

Note:

- Here, $U(c_0, c_1, \dots, c_\infty)$ is assumed to be time separable: it is a discounted sum of period utilities. It is possible to deviate from time separability: a famous example is Epstein-Zin type of recursive utility

$$U_t \equiv U(c_t, c_{t+1}, \dots) = [(1 - \beta)c_t^\rho + \beta U_{t+1}^\rho]^{1/\rho}$$

there are midterm/comp questions that use this utility function (Jan 2011 comp, 2009 Fall Midterm Q1).

- (Optional) Some technical conditions to impose on $u(\cdot)$:
 - (Stokey and Lucas, Chapter 4, Assumption 4.2) Assume that for all initial asset holding a_0 and $\{(c_t, a_{t+1})\}_{t=0}^{\infty}$ satisfying the budget constraints (1), $\lim_{T \rightarrow \infty} \sum_{t=0}^T \beta^t u(c_t)$ exists.
 - We don't know if there is an economic meaning for negative consumption, and hence we typically assume that $c_t \geq 0$. This essentially introduces an additional “constraint” on consumption. To circumvent this issue, we usually introduce an *Inada condition* on $u(\cdot)$: namely, we assume that

$$\lim_{c \rightarrow 0} u'(c) = \infty \text{ and } \lim_{c \rightarrow \infty} u'(c) = 0$$

With Inada condition, choosing to consume $c = 0$ is never optimal because the marginal utility of consumption is infinity (so that increasing consumption will always make the consumer better off). Similarly, having an explosive consumption $c = \infty$ is also not optimal because the marginal utility is 0 (so that reducing consumption will always make the consumer better off).

The Consumption-saving Problem The consumption-saving problem is

$$\begin{aligned} \max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t. } \quad & \forall t, c_t + a_{t+1} = (1+r)a_t + y_t \end{aligned} \tag{3}$$

with the initial wealth a_0 taken as given.

1.2 Solving Consumption-saving Dynamics

There are multiple ways of solving (3), a standard dynamic programming problem.

Method 1: Working with IBC

One way of solving (3) is to combine the period budget constraints (1) into one single, intertemporal budget constraint (IBC).

Stack the period budget constraints from period 0:

$$\begin{aligned} \frac{a_1}{1+r} &= a_0 + \frac{y_0 - c_0}{1+r} \\ \frac{a_2}{(1+r)^2} &= \frac{a_1}{1+r} + \frac{y_1 - c_1}{(1+r)^2} \\ &\dots \\ \frac{a_T}{(1+r)^T} &= \frac{a_{T-1}}{(1+r)^{T-1}} + \frac{y_{T-1} - c_{T-1}}{(1+r)^T} \end{aligned}$$

Sum them up:

$$\frac{a_T}{(1+r)^T} = a_0 + \frac{1}{1+r} \sum_{t=0}^{T-1} \frac{1}{(1+r)^t} (y_t - c_t) \tag{4}$$

If we impose a no-Ponzi condition

$$\lim_{T \rightarrow \infty} \frac{a_T}{(1+r)^T} = 0 \tag{5}$$

then take limits $T \rightarrow \infty$ on both sides of (4) and plug (5) in:

$$0 = a_0 + \frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} (y_t - c_t)$$

or, equivalently,

$$\frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} c_t = a_0 + \frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} y_t \quad (6)$$

Equation (6) is called Intertemporal Budget Constraint (IBC):

- LHS is the present value of lifetime expenditure;
- RHS is the present value of lifetime income/resources.
- IBC is a function of $\{c_t\}$, a_0 and $\{y_t\}$ only; a_{t+1} terms are canceled out in the summation (hence, no longer need to include them in the control).

Now we can rewrite (3) as

$$\begin{aligned} & \max_{\{c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t. } & \frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} c_t = a_0 + \frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} y_t \end{aligned} \quad (7)$$

This modified problem can be directly solved by the standard Lagrangian multiplier method with one single constraint (6):

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) - \lambda \left(\frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} c_t - a_0 - \frac{1}{1+r} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} y_t \right)$$

Then take first-order conditions on c_t :

$$\beta^t u'(c_t) = \lambda \frac{1}{(1+r)^{t+1}} \quad (8)$$

The optimal consumption path $\{c_t\}$ and Lagrange multiplier λ satisfies (8) and (6).

If we write (8) for c_{t+1} :

$$\beta^{t+1} u'(c_{t+1}) = \lambda \frac{1}{(1+r)^{t+2}}$$

and divide it by (8), we get

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = 1 + r \quad (9)$$

This is called “Euler equation”. It gives us the tradeoff between today and tomorrow’s consumption.

Method 2: Lagrange Multiplier

The second way of solving (3) is to directly apply Lagrange multiplier method to (3). It’s a constrained optimization problem with infinitely many controls and constraints. The Lagrangian becomes

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \{u(c_t) - \lambda_t (c_t + a_{t+1} - y_t - (1+r)a_t)\}$$

and in each period, the consumer chooses c_t and a_{t+1} to maximize $\mathcal{L}$. The first order condition on c_t is given by

$$u'(c_t) = \lambda_t \tag{10}$$

the first order condition on a_{t+1} is given by

$$-\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} (1+r) = 0 \tag{11}$$

and the first order condition on λ_t is simply the period budget constraint (1). The optimal consumption path $\{c_t\}$, asset holding $\{a_t\}$ and Lagrange multiplier $\{\lambda_t\}$ is therefore given by (10), (11), (1) plus a transversality condition

$$\lim_{t \rightarrow \infty} \beta^t \lambda_t a_{t+1} = 0$$

If we combine (10) and (11), we get

$$u'(c_t) = \beta(1+r)u'(c_{t+1}) \tag{12}$$

which is the same Euler equation as (9).

Method 3: Bellman Equation

In this course, the recommended way of solving a dynamic programming problem is to use Bellman equation. In the exams (and later when you are writing macro papers), you will need to formulate macroeconomic questions as a dynamic programming problem and write down the Bellman equation correctly.

No matter what problems you are facing, the first step is always to identify *control* and

state variables. Control variables are variables that can be freely chosen at period t . State variables are variables that are predetermined in period t and affect future objectives or constraints. If the value of state variable in the future is affected by today's control variable, then we call such state variables "endogenous states"; else, if the value of state variables in the future is independent of today's control variable, we call such variables "exogenous states".

For instance, in the consumption-saving model here:

- Control variable: consumption c_t ;
- Endogenous state variable: asset holding a_t [it is a stock variable and predetermined in period $t - 1$, so that the consumer takes it as given when making decision in period t ; also, the choice of c_t affects asset holding at the beginning of next period a_{t+1} , hence it is endogenous];
- Exogenous state variable: y_t and r [these are exogenous in the current problem and is taken as given for the consumer; choice of consumption c_t does not affect the value of r and y_t , therefore it is exogenous].

When the problem gets more complicated, it is more important to specify state and control variables correctly. A useful exercise now is to go to past exam/comp questions and see if you can write down state and control variables correctly from the verbal description in the questions.

Next, we define value function and policy function. Value function is a function $v : a \mapsto \mathbb{R}$ that maps states into a real number. Define the value function $v(a)$ as

$$v(a) \equiv \max_{\{c_t, a_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t) \tag{13}$$

s.t. $c_t + a_{t+1} = (1 + r)a_t + y_t$
and $a_0 = a$ is taken as given.

In other words, value function $v(a)$ specifies the maximum lifetime utility that the consumer could get if he starts with asset holding a . Policy function is a mapping from state space to action/control space; it specifies a decision rule (in this example: policy function $c(a)$ specifies how much to consume if my asset holding is a).

The next step is to write (13) recursively. Note that

$$v(a) = \max u(c_0) + \underbrace{\beta \sum_{t=0}^{\infty} \beta^t u(c_{t+1})}_{\equiv v(a_1)}$$

therefore, value function $v(a)$ and a policy function $c(a)$ should solve

$$\begin{aligned} v(a) &= \max_c u(c) + \beta v(a') \\ \text{s.t. } a' &= (1+r)a + y - c \end{aligned} \tag{14}$$

We call (14) “Bellman equation”.

The solution to Bellman equation involves a first-order condition on control variable c :

$$u'(c) - \beta v'(a') = 0 \tag{15}$$

plus an envelope condition

$$v'(a) = \beta(1+r)v'(a') \tag{16}$$

which implies

$$v'(a) = (1+r)u'(c)$$

and

$$v'(a') = (1+r)u'(c')$$

and therefore

$$u'(c) = \beta(1+r)u'(c') \tag{17}$$

again, we end up with the same Euler equation as in the other two methods.

Notes:

- Analytical solutions are generally unavailable - need numerical solution;
- Some technical aspects to think about:
 - Is the solution to Bellman equation (14) unique?
 - Under what conditions are solutions to Bellman equation (14) equivalent to the solutions given by the other two methods?
 - If (14) needs to be solved numerically, does the algorithm converge?

The answers to these questions are in Chapter 3 and 4 of Stockey and Lucas. We will return to this in Professor Aruoba's part.

1.3 Examples

[To be added]

1.4 Liquidity Constraints

So far we've been focusing on problems with no borrowing or liquidity constraints. This means that asset holding a_t can take any real numbers.

Now suppose we have a borrowing constraint

$$a_{t+1} \geq 0 \text{ for all } t \geq 0 \quad (18)$$

i.e. the consumer is not allowed to borrow at all.

Note:

- The formulation (18) is largely without loss of generality: if the borrowing constraint looks like $a_{t+1} \geq \underline{a}$ for all t , then we can always redefine $\tilde{a}_t \equiv a_t - \underline{a}$ and treat $\tilde{a}_t$ as “asset holding”.
- (Optional) However, if we do not introduce assumptions on $u(\cdot)$ that guarantee an interior solution (e.g. Inada condition, discussed above), then the consumption being non-negative $c_t \geq 0$ restricts the choice of a_t in each period. In period t , when choosing a_{t+1} , the consumer knows that her future consumption $c_{t+1}, c_{t+2}, \dots$ will all be non-negative, and therefore

$$\begin{aligned} a_{t+2} &= a_{t+1}(1+r) + y_{t+1} - c_{t+1} \leq a_{t+1}(1+r) + y_{t+1} \\ a_{t+3} &= a_{t+2}(1+r) + y_{t+2} - c_{t+2} \leq a_{t+2}(1+r) + y_{t+2} \\ &\dots \end{aligned}$$

and hence

$$\begin{aligned} \frac{a_{t+2}}{1+r} &\leq a_{t+1} + \frac{y_{t+1}}{1+r} \\ \frac{a_{t+3}}{(1+r)^2} &\leq \frac{a_{t+2}}{1+r} + \frac{y_{t+2}}{(1+r)^2} \\ &\dots \end{aligned}$$

Summing these inequality up and imposing the no-Ponzi condition, we get

$$0 \leq a_{t+1} + \sum_{s=1}^{\infty} \frac{y_{t+s}}{(1+r)^s}$$

or equivalently,

$$a_{t+1} \geq - \sum_{s=1}^{\infty} \frac{y_{t+s}}{(1+r)^s}$$

This constraint is called “natural borrowing constraint” in the literature. Will be discussed in the lecture.

How would this change the three methods we discussed above? The mathematical tool we use is Kuhn-Tucker condition: basically, the borrowing constraints introduce a complementary-slackness condition in each period.

Let μ_t be the Lagrange multiplier for these slackness conditions. We simply augment them into the three methods we have above.

For example, if you used method 2, then the Lagrangian becomes

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \{u(c_t) - \lambda_t (c_t + a_{t+1} - y_t - (1+r)a_t) + \mu_t a_{t+1}\}$$

Now the first order condition becomes

$$u'(c_t) = \lambda_t \tag{19}$$

and

$$\beta^t (-\lambda_t + \mu_t) + \beta^{t+1} (1+r) \lambda_{t+1} = 0 \tag{20}$$

plus a slackness conditions for μ_t :

$$\mu_t \geq 0 \text{ with equality hold when } a_{t+1} > 0 \tag{21}$$

Combining the three optimality conditions above, we get

$$u'(c_t) - \mu_t = \beta(1+r)u'(c_{t+1})$$

or equivalently,

$$u'(c_t) \geq \beta(1+r)u'(c_{t+1}) \text{ with equality hold when } a_{t+1} > 0 \tag{22}$$

Several takeaways from (22):

- If the consumer is a saver, then she will choose $a_{t+1} > 0$; in this case, the borrowing constraint is not binding, and the Euler equation continues to hold with equality;
- If the consumer is a borrower, i.e. absent the borrowing constraint she would have chosen $a_{t+1} < 0$, then her consumption behavior will be changed by the borrowing constraint: now, she is not allowed to borrow, and the best she can do is to consume all her resources in period t and end up with $a_{t+1} = 0$ — in other words, the borrowing constraint is binding.

Illustration with a Two-period Model

It's probably easier to illustrate the intuition in a two-period model (in an intermediate-macro fashion).

Setup Let $t \in \{1, 2\}$. For simplicity, assume that the initial asset holding $a_1 = 0$. The consumer faces budget constraints

$$c_1 + a_2 = y_1 \tag{23}$$

$$c_2 = y_2 + (1 + r)a_2 \tag{24}$$

and she chooses consumption plan (c_1, c_2) and saving a_2 to maximize her lifetime utility

$$u(c_1) + \beta u(c_2) \tag{25}$$

Note: This two-period problem is extremely useful in illustrating the intuition. Besides, it allows for a flexible economic interpretation, making it applicable to many macroeconomic problems. For instance, one may think of $t = 1$ as a young cohort and $t = 2$ as an old cohort, and you may build an economy with a mixture of old and young cohort with each agents living only for two periods — this is called “overlapping generation model (OLG)”.

A Graphical Illustration

I could derive IBC in the same manner as before:

$$c_1 + \frac{c_2}{1 + r} = y_1 + \frac{y_2}{1 + r} \tag{26}$$

and we can plot it on a (c_1, c_2) plane. The slope of IBC is given by $(1 + r)$: it is the market exchange rate between today and tomorrow's consumption goods (i.e. the market compensates the consumer $(1 + r)$ units of consumption goods tomorrow in order to persuade her to give up 1 unit of consumption good today).

On the same graph, we can plot the indifference curve from the lifetime utility function. The slope of indifference curve is given by $\frac{u'(c_1)}{\beta u'(c_2)}$, which is the marginal rate of substitution between today and tomorrow's consumption.

At optimum, the budget constraint and the indifference curve are tangent to each other. In microeconomics this is simply “MRS = price”; in this macroeconomic setting, it is the Euler equation that we derived before.

Now, with borrowing constraint $a_2 \geq 0$, it basically means that

$$c_1 \leq y_1 \text{ and } c_2 \geq y_2$$

This is essentially introducing a kink to the intertemporal budget constraint: only the part of the budget set on the left of $c_1 = y_1$ can be chosen.

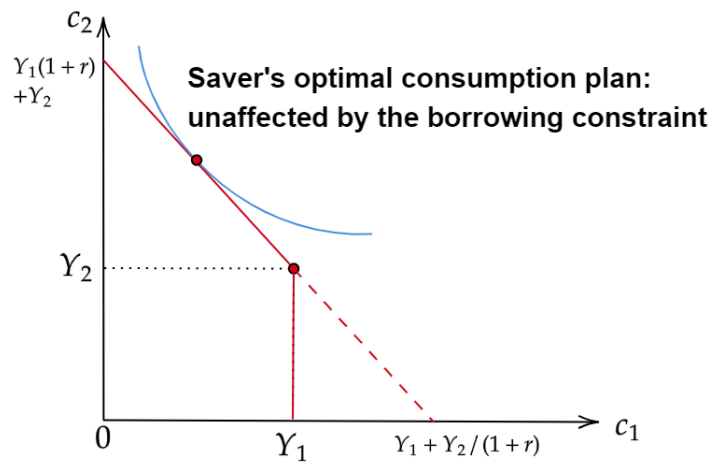


Figure 1: Saver's optimal plan is not affected by the borrowing constraint

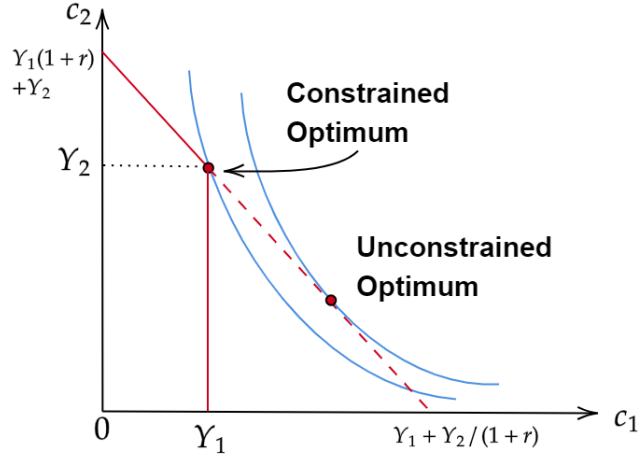


Figure 2: Borrower's optimal plan is to consume the endowment in each period.

2 Consumption Dynamics under Uncertainty

Now, we incorporate uncertainty in the dynamic programming problem. In the certainty problem, what we have done is to solve an optimal consumption and saving path for a given income path. Now, in the uncertainty problem, at period 0 the consumer no longer knows the realization of the lifetime income path; instead, what she knows in period 0 is only y_0 but not the future path $y_1, y_2, \dots$. Similarly, in each period t , she knows only the realization of all past income $y_0, y_1, \dots, y_t$, but not the realization of future income path $y_{t+1}, y_{t+2}, \dots$.

In the literature, we often use $y^t \equiv \{y_0, y_1, \dots, y_t\}$ to denote the past realizations of income on and before period t , or, the *history* of t . The history y^t can also be recursively written as

$$y^{t+1} = y^t \cup \{y_{t+1}\}$$

and sometimes it reads “history y^{t+1} is a continuation of history y^t ”.

Same as before, assume that the consumer can save or borrow with a risk-free bond at interest rate r . The consumer's period budget constraint remains the same:

$$a_{t+1} = (1 + r)a_t + y_t - c_t$$

Keep in mind that the c_t and a_{t+1} I write here is a mathematically different object from the certainty case. Here, c_t and a_{t+1} stand for consumption and asset holding at **history** y^t . In other words, the period budget constraint should hold for any realized path of income. Hence, a more precise notation should be $c_t(y^t)$ and $a_{t+1}(y^t)$ for c_t and a_{t+1} , and the budget

constraint should be more precisely written as

$$a_{t+1}(y^t) = (1 + r)a_t(y^{t-1}) + y_t - c_t(y^t)$$

It is a convention in the macro literature to simplify the notation $c_t(y^t)$ to c_t , although some economists (mostly from a Minnesota background) still keep the notation $c_t(y^t)$ (e.g. Chapter 7 of LS, Chari and Kehoe).

With uncertainty, from period-0 perspective, the consumer does not know the realization of future income path, and she needs to make a **plan** $\{c_t(y^t), a_{t+1}(y^t)\}_{t=0}^{\infty}$ where $c_t(y^t)$ and $a_{t+1}(y^t)$ defines the consumption and saving at period t when the realized income path up to period t is y^t . The stochastic dynamic programming problem can be written as

$$\begin{aligned} \max_{\{c_t(y^t), a_{t+1}(y^t)\}_{t=0}^{\infty}} \quad & \sum_{y^t} \pi(y^t) \sum_{t=0}^{\infty} \beta^t u(c_t(y^t)) \\ \text{s.t. } \quad & \forall t \geq 0 \text{ and } y^t, a_{t+1}(y^t) = (1 + r)a_t(y^{t-1}) + y_t - c_t(y^t) \end{aligned} \quad (27)$$

where $\pi(y^t)$ is the unconditional probability of income path y^t .

Let's stare at the objective function. The discounted sum part $\sum_{t=0}^{\infty} \beta^t u(c_t(y^t))$ should look familiar to us: it is just the lifetime utility of consumption plan $\{c_t(y^t)\}$ for a given income path y^t , the same one as in the certainty problem. The new part is $\sum_{y^t} \pi(y^t)$: it simply captures the fact that at period 0, the consumer only knows the realization of y_0 but does not know $y_1, y_2, \dots$ so that when he is evaluating consumption and saving plans, he needs to take expectation with respect to all the possible realization of income path in the future.

It is the convention of the macro literature to write $\sum_{y^t} \pi(y^t)$ as $\mathbb{E}_0$ (call it "time-0 expectation operator"), and $\sum_{y^{t+s}|y^t} \pi(y^{t+s}|y^t)$ as $\mathbb{E}_t$ ("time- t expectation operator").¹ Hence, the above stochastic dynamic programming problem is usually written as

$$\begin{aligned} \max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t. } \quad & \forall t \geq 0, a_{t+1} = (1 + r)a_t + y_t - c_t \end{aligned} \quad (28)$$

¹Here, $y^{t+s}|y^t$ means y^{t+s} is a continuation of y^t , and $\pi(y^{t+s}|y^t)$ is the probability of history y^{t+s} conditional on history y^t .

2.1 Markov Decision Process

In period t , the consumer does not know the realization of y_{t+1} (and $y_{t+2}, y_{t+3}, \dots$). In order to make consumption-saving decisions in period t , the consumer needs to form expectation about future income y_{t+1} . Mathematically, the consumer needs to know the transition probability $P(y_{t+1}|y^t)$, which gives us the probability of the income being y_{t+1} tomorrow if the experienced path of income is y^t .

We know that y^t is a high-dimensional object and the transition probability $P(y_{t+1}|y^t)$ is time-varying. Hence, it is difficult to present the transition probability in a handy form for analysis or computation (e.g. a matrix). A usual simplification made by the macroeconomics is to assume that the transition probability $P(y_{t+1}|y^t)$ is *Markov*:

$$P(y_{t+1}|y^t) = P(y_{t+1}|y_t) \quad (29)$$

In other words, the probability of next-period income being at y_{t+1} only depends on today's realized income y_t and is independent of earlier realizations of income $y_{t-1}, y_{t-2}, \dots, y_0$. The consumer only needs to know her current income y_t to forecast her future income $y_{t+1}, \dots$.

It turns out that this greatly simplifies the stochastic dynamic programming problem. With the Markov property, it turns out that the optimal consumption and saving plan $c_t(y^t)$ and $a_{t+1}(y^t)$ will be independent of past realizations $y_{t-1}, y_{t-2}, \dots$, and depend only on the current income y_t ; in other words, with Markov property, the policy function at each history y^t will be $c(y_t)$. This will be discussed in the lecture and revisited in Boragan's part.

2.2 Bellman Formulation (Briefly) and Relationship with Exams

In the lecture, we have seen (or will see) the method of using Bellman equation to solve a dynamic programming problem. This involves finding in a functional space a value function that maps states to the maximum expected discounted sum of utilities, and a policy function or a contingency plan that maps states to a decision.

With the Markov property above, we know that in each period t ,

- control variable: c_t, a_{t+1} ;
- endogenous state variable: a_t ;
- exogenous state variable: y_t (if we don't have Markov property, this should be the history y^t).

The stochastic dynamic programming problem can be written as a Bellman equation

$$\begin{aligned} v(a; y) &= \max_{c, a'} u(c) + \beta \mathbb{E}[v(a'; y')|y] \\ \text{s.t. } \forall t, c + a' &= (1 + r)a + y \end{aligned} \quad (30)$$

Replace a' in the expectation by the budget constraint:

$$v(a; y) = \max_c u(c) + \beta \mathbb{E}[v((1 + r)a + y - c; y')|y]$$

First order condition on c :

$$u'(c) - \beta \mathbb{E}[v_1((1 + r)a + y - c; y')|y] = 0 \quad (31)$$

where v_1 denotes the partial derivative of v with respect to the first argument (endogenous state a).

The envelope condition is

$$v_1(a; y) = \beta \mathbb{E}[v_1((1 + r)a + y - c; y')|y] \quad (32)$$

Equation (31) and (32) jointly define the optimality conditions that a solution to Bellman equation (30).

Combine (31) and (32), we get

$$u'(c) = v_1(a; y)$$

which implies that

$$u'(c') = v_1(a'; y')$$

and hence (31) becomes

$$u'(c) = \beta(1 + r)\mathbb{E}[u'(c')|y] \quad (33)$$

This is the Euler equation for the stochastic dynamic programming problem.

Note:

- The way I write down the value function may be slightly different from what's being taught in the lecture. Although we do not need envelope conditions for exogenous variable y in our case, I still include it as an argument of the value function, because in general the value functions of endogenous state a evaluated at different exogenous states y 's are different. This is because if we are at different exogenous states, the transition probability $P(y'|y)$ is different, so that $\mathbb{E}[\cdot|y]$ can be different if we have different

realization of current state y . Later you will learn that including exogenous states helps with both computationally solving (30) and incorporating general equilibrium forces (such as prices) in Boragan's part.

A Classification of Past Exam Questions

All the exam questions in this part can be regarded as adding “tweaks” to (30). I prefer to classify the exam questions by the types of tweaks in the question:

1. Tweaks on $\mathbb{E}$: incorporating non-standard belief/expectation (a fancier term is “behavioral macroeconomics”) - increasingly popular in recent year's exams (2022 June Comp, 2023 Fall Midterm ,2024 June Comp);
2. Tweaks on state variables: the most commonly seen question is to change a to $\vec{a}$: instead of having only one endogenous variables, we may incorporate multiple types of assets (e.g. risky vs risk-free asset, liquid vs illiquid, housing, durable goods, etc.);
3. Tweaks on control variables: again, one could change c to $\vec{c}$: we may have a bundle of different types of consumption goods to choose. The most common ones are the questions with multiple sectors (e.g. tradeable vs non-tradeable goods, manufacturing goods vs service goods, etc.)
4. Tweaks on utility function: we may have non time-separable utility;
5. Non household problems: we may have firm's problem or an entrepreneur's problem in the question;
6. Questions with liquidity constraints (we've seen them already).

In the discussion sessions that follow, I will discuss one genre of tweaks each time.

2.3 Sidenote: Lagrangian Method

I now sketch how to solve (27) by Lagrangian method

Set up the Lagrangian:

$$\mathcal{L} = \sum_{y^t} \pi(y^t) \sum_{t=0}^{\infty} \beta^t \{ u(c_t(y^t)) - \lambda_t(y^t) (a_{t+1}(y^t) + c_t(y^t) - (1+r)a_t(y^{t-1}) - y_t) \} \quad (34)$$

FOC on $c_t(y^t)$:

$$u'(c_t(y^t)) = \lambda_t(y^t) \quad (35)$$

or, using the simplification,

$$u'(c_t) = \lambda_t \quad (36)$$

FOC on $a_{t+1}(y^t)$ is a bit trickier. First, $a_{t+1}(y^t)$ shows up in history- y^t budget constraint. Second, $a_{t+1}(s^t)$ shows up in all history- y^{t+1} budget constraint if y^t is a continuation of y^t . I collect the terms in the Lagrangian with $a_{t+1}(s^t)$:

$$\pi(y^t)\beta^t \{-\lambda_t(y^t)a_{t+1}(y^t)\} + \sum_{y^{t+1}|y^t} \pi(y^{t+1})\beta^{t+1}\lambda_{t+1}(y^{t+1})(1+r)a_{t+1}(y^t)$$

Take FOC on $a_{t+1}(y^t)$:

$$\begin{aligned} \pi(y^t)\beta^t\lambda_t(y^t) &= \beta^{t+1}(1+r) \sum_{y^{t+1}|y^t} \pi(y^{t+1})\lambda_{t+1}(y^{t+1}) \\ \implies \pi(y^t)\lambda_t(y^t) &= \beta(1+r) \sum_{y^{t+1}|y^t} \pi(y^{t+1})\lambda_{t+1}(y^{t+1}) \\ \implies \lambda_t(y^t) &= \beta(1+r) \sum_{y^{t+1}|y^t} \underbrace{\frac{\pi(y^{t+1})}{\pi(y^t)}}_{\equiv \pi(y^{t+1}|y^t)} \lambda_{t+1}(y^{t+1}) \end{aligned} \quad (37)$$

Using our previous definition of $\mathbb{E}_t$ and simplify $\lambda_t(y^t)$ as λ_t , we get

$$\lambda_t = \beta(1+r)\mathbb{E}_t[\lambda_{t+1}] \quad (38)$$

Hence, we get Euler equation:

$$u'(c_t) = \beta(1+r)\mathbb{E}_t[u'(c_{t+1})] \quad (39)$$

It's the same Euler equation that we got with Bellman method.